

Notation for American FX option in presence of an Extended Vasicek interest rate model

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Abstract

We briefly summarize the problem, the main equations involved and the suggested discretization method.

1 Notation and Basic Definitions

Let us consider S_t as the EUR-USD Foreign Exchange, i.e. the number of US Dollars needed to buy one euro for an US investor. The underlying dynamics for a Foreign Exchange, e.g. the EUR-USD, is

$$dS_t = S_t [(r_t - d_t)dt + \sigma_S dW_t]$$

with initial condition S_0 and d_t the interest rate in the foreign country (EUR in this case), a deterministic function of time.

The USD interest rate r_t is stochastic and follows an Extended Vasicek model (known also as Hull-White model)

$$r_t \equiv \hat{\zeta}_t + \varphi_t$$

with φ_t a deterministic function of time and

$$d\hat{\zeta}_t = -\kappa \hat{\zeta}_t dt + \sigma_r dz_t$$

where

$$dz_t dW_t = \rho dt \quad .$$

Having introduced the Cost-of-carry function as $c_t \equiv \varphi_t - d_t$, we can define the

$$\left\{ \begin{array}{l} \xi_t \equiv \frac{1}{\sigma_S} \left\{ \ln \frac{S_t}{S_0} - \int_{t_0}^t \left(c_u - \frac{\sigma_S^2}{2} \right) du \right\} \\ \zeta_t \equiv \frac{\hat{\zeta}_t}{\sigma_r} \end{array} \right.$$

which follow the SDEs

$$\left\{ \begin{array}{l} d\xi_t = \frac{\sigma_r}{\sigma_S} \zeta_t dt + dW_t \\ d\zeta_t = -\kappa \zeta_t dt + dz_t \end{array} \right. \quad . \quad (1)$$

As in [1] we can define the variable χ_t orthogonal to ζ_t s.t.

$$d\chi_t \equiv \frac{d\zeta_t - \rho d\zeta_t}{\sqrt{1 - \rho^2}}$$

and this allows rewriting (1). The set of SDEs becomes

$$\begin{cases} d\chi_t &= \theta \zeta_t dt + dv_t \\ d\zeta_t &= -\kappa \zeta_t dt + dz_t \end{cases} \quad (2)$$

with initial conditions $\chi_0 = \zeta_0 = 0$.

We have defined

$$\theta \equiv \frac{1}{\sqrt{1 - \rho^2}} \left(\frac{\sigma_r}{\sigma_s} - \rho \kappa \right)$$

and

$$dv_t \equiv \frac{dW_t - \rho dz_t}{\sqrt{1 - \rho^2}}$$

s.t. $dv_t dz_t = 0$.

2 Calibration: Relations with zero-rates curves

The Discount factor between s and $t \geq s$ is

$$B(s, t) = E_s[D(s, t)] = \exp \left\{ - \int_s^t \varphi_u du - \zeta_s \sigma(s, t) + \frac{1}{2} \int_s^t \sigma(u, t)^2 du \right\}$$

where

$$\sigma(u, t) \equiv \sigma_r \frac{1 - \exp[-\kappa(t - u)]}{\kappa}$$

is the associated Gaussian HJM volatility to the US interest rate.

We have then that the USD initial discount curve is

$$B(t_0, t) = \exp \left\{ - \int_{t_0}^t \varphi_u du + \frac{1}{2} \int_{t_0}^t \sigma(u, t)^2 du \right\}$$

while for the EUR case

$$B_{\text{€}}(t_0, t) = \exp \left\{ - \int_{t_0}^t d_u du \right\} .$$

and FX forward rate in t_0 with delivery in t is

$$F(t_0, t) = S_0 \frac{B_{\text{€}}(t_0, t)}{B(t_0, t)} .$$

It can be useful to remind that

$$\mathcal{R}(s, t) \equiv \int_s^t r_u du = \int_s^t \varphi_u du + \zeta_s \sigma(s, t) + \int_s^t \sigma(u, t) dz_u \quad .$$

3 Pricing: Discretization and binomial tree

The key step for pricing derivative instruments is to provide a discrete approximations for χ_t and ζ_t in equation (2). In order to construct the (binomial) tree, following the notation in [2], one divides the time to maturity T into N intervals of length $h = T/N$, setting the size of each step equal to $\sqrt{h}$, since both processes are of unit variance. At time $t = i h$ ($i = 1, \dots, N$), χ may be located at one of the nodes (i, k) ($k = 1, \dots, i$) while ζ may be located at one of the nodes (i, j) ($j = 1, \dots, i$) corresponding to the values

$$\begin{aligned} \chi_{i,k} &= (2k - i)\sqrt{h} \\ \zeta_{i,j} &= (2j - i)\sqrt{h} \end{aligned}$$

Then it is possible to follow the same steps described either in [1] or in [2] (censored technique).

References

- [1] J. Z. Wei, *Valuing American Equity option with a Stochastic Interest Rate: A Note*, The Journal of Financial Engineering **2** (1996), 195-206.
- [2] E. Appolloni, L. Caramellino and A. Zanette, *A robust tree method for pricing American options with the CoxIngersollRoss interest rate model*, IMA Journal of Management Mathematics, (2014) 1-25.